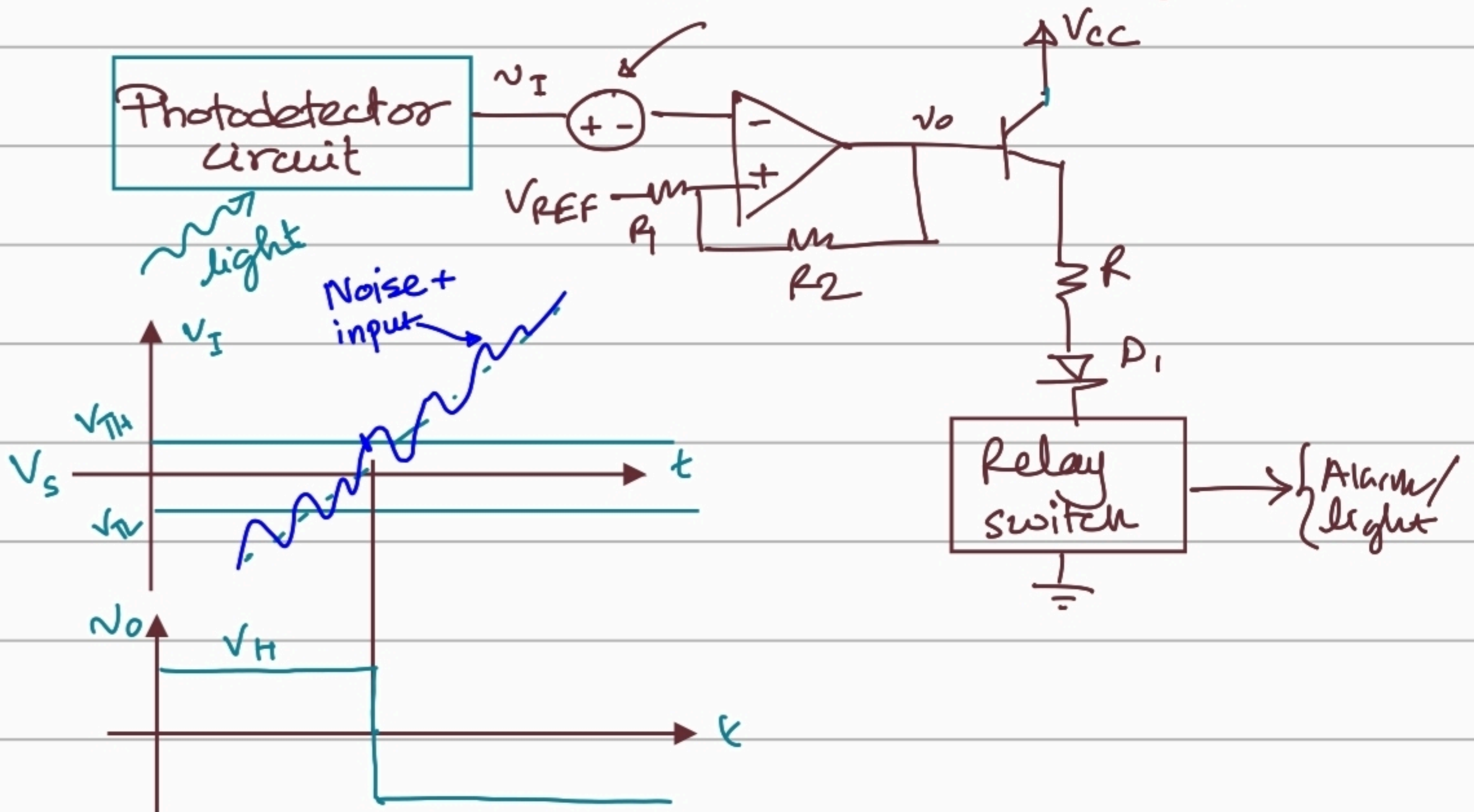


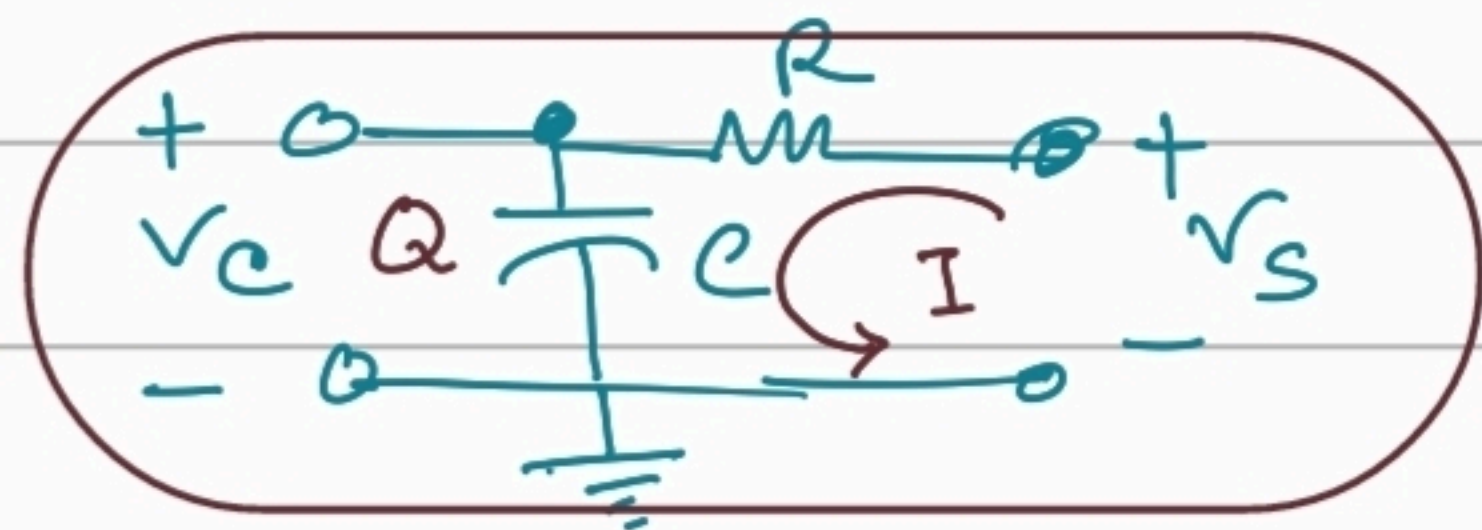
Noise source: Chattering effect



Design problem 15.7 (Practice | Final Exam)

Lecture 19: Signal Generators.

Basic R-C circuit:



Suppose, at initial time t_1 , the capacitor has a voltage $V_C(t_1) = V_{\text{initial}}$.

If $V_S > V_C(t_1)$, $Q \uparrow$ and

if $V_S < V_C(t_1)$, $Q \downarrow$.

From the definition of current flow, we know that, $I = \frac{dQ}{dt}$

Using the definition of capacitance ($C = \frac{Q}{V}$)

$$I = \frac{dQ}{dt} = \frac{d}{dt}(CV_C) = \boxed{C \frac{dV_C}{dt} = \frac{V_S - V_C}{R}}$$

We want to determine the voltage across capacitor for the time $t_1 < t < \infty$.

$$C \frac{dV_c}{dt} = \frac{V_s - V_c}{R} \Rightarrow \int_{V_c(t_1)}^{V_c(t)} \frac{dV_c}{V_s - V_c} = \int_{t_1}^t \frac{dt}{RC}$$

$$\Rightarrow -\ln|V_s - V_c| \Big|_{V_c(t_1)}^{V_c(t)} = \frac{t - t_1}{RC} \Rightarrow t - t_1 = RC \ln \left| \frac{V_s - V_c(t_1)}{V_s - V_c(t)} \right|$$

$$\Rightarrow \boxed{V_c(t) = V_s + (V_c(t_1) - V_s) \exp\left(-\frac{t - t_1}{RC}\right)}$$

$t \rightarrow \infty$, $V_c(\infty) = V_s$ because $\exp(-\infty) \rightarrow 0$.

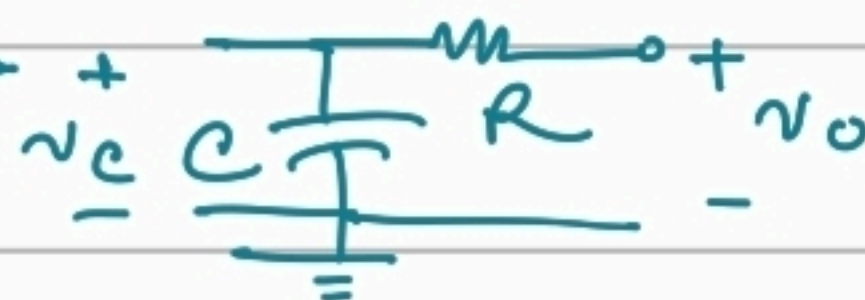
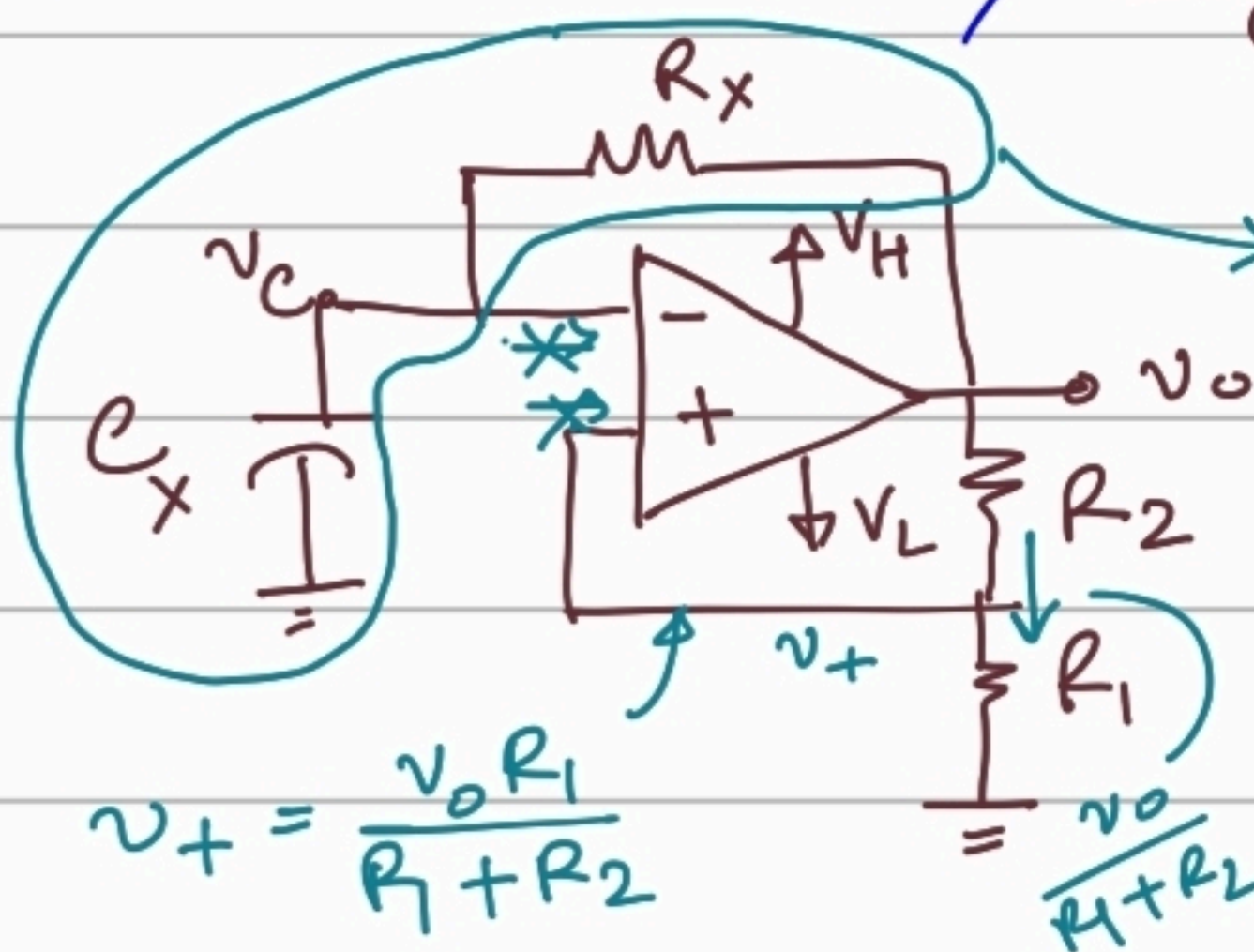
$V_c(\infty) = V_s = V_{\text{final}}$ and $V_c(t_1) = V_{\text{initial}}$.

$\tau = RC = \text{time constant}$.

$$\boxed{V_c(t) = V_{\text{final}} + (V_{\text{initial}} - V_{\text{final}}) \exp\left(-\frac{t - t_1}{\tau}\right)}$$

Schmitt Oscillator / Astable Multivibrator:

There is no stable state.



$$V_c(t) = V_O + (V_{\text{initial}} - V_O) \exp\left(-\frac{t - t_1}{R_X C_X}\right)$$

$$t - t_1 = \tau \ln \left| \frac{V_{\text{final}} - V_c(t_1)}{V_{\text{final}} - V_c(t_2)} \right|$$

We already know the threshold voltages.

$$V_{TH} = V_H \frac{R_1}{R_1 + R_2} \quad \text{and} \quad V_{TL} = V_L \frac{R_1}{R_1 + R_2}$$

Coughline. Chapter 6.

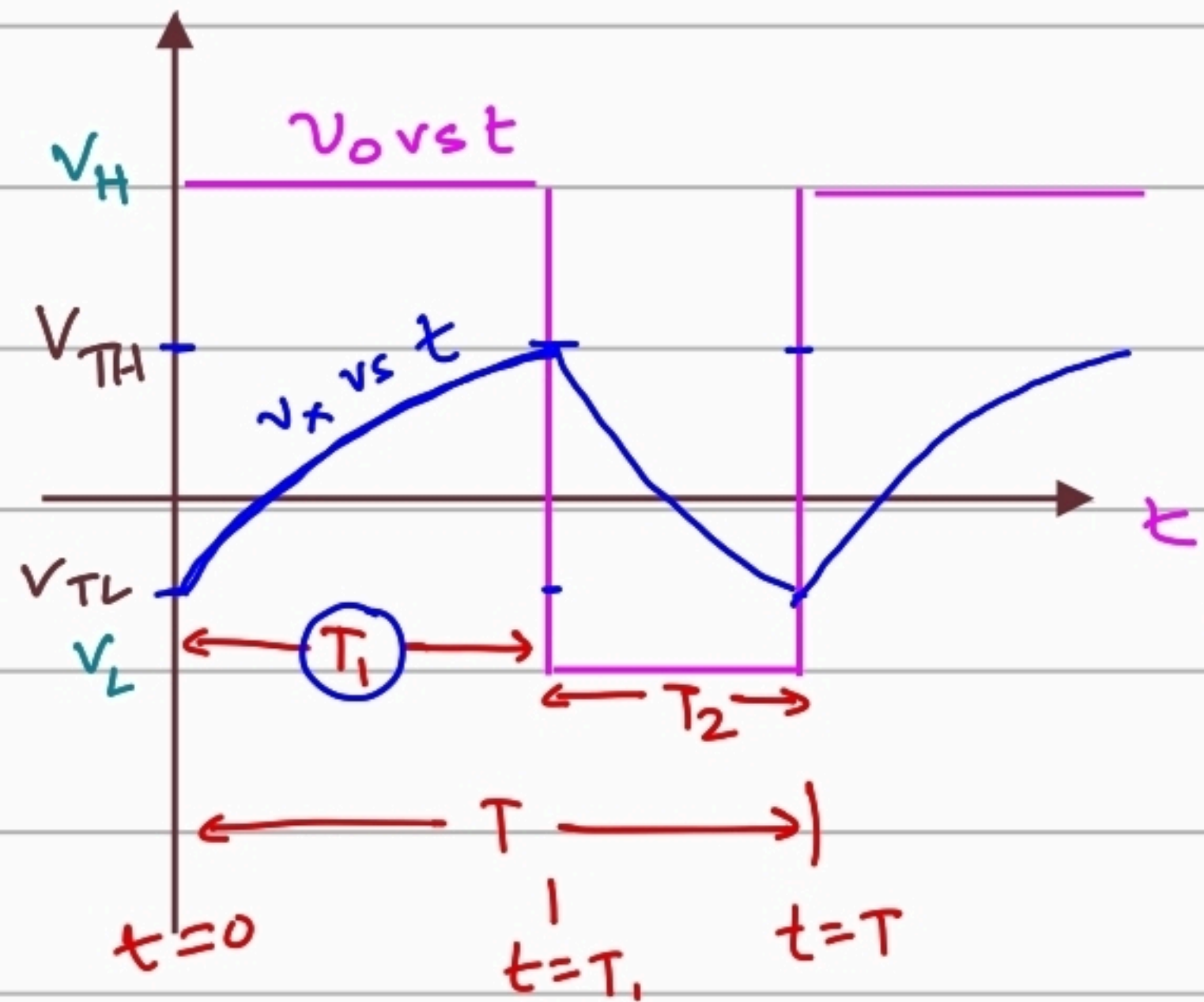
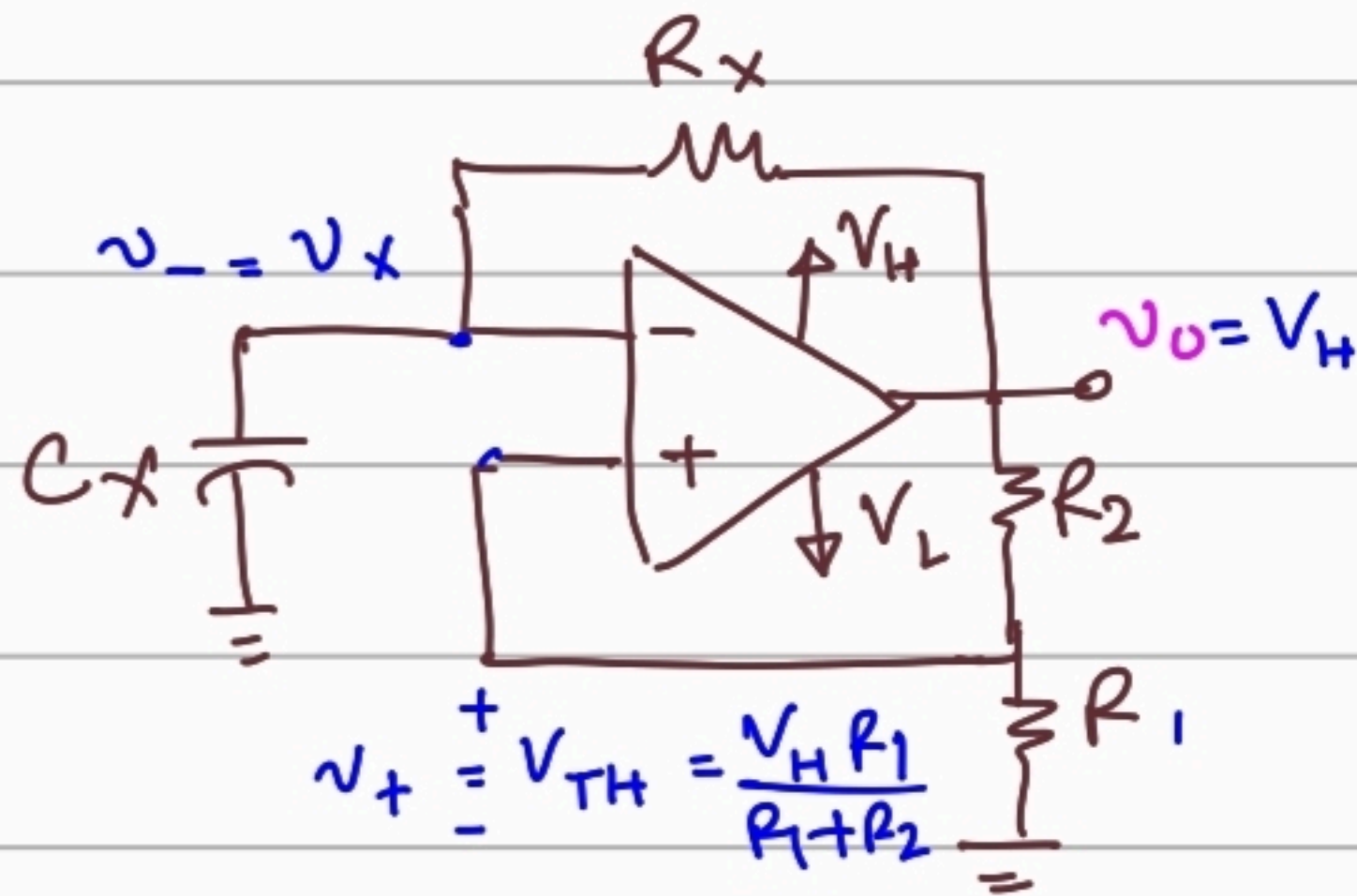
$V_{TH} \rightarrow V_{UT} = \text{upper threshold voltage}$

$V_{TL} \rightarrow V_{LT} = \text{lower threshold voltage}$

$V_H \rightarrow V_{\text{sat}}$, $V_L \rightarrow -V_{\text{sat}}$, $|V_{\text{sat}}| \neq |-V_{\text{sat}}|$

Square Wave Generator: (steady state analysis)

① Case 1: $0 < t < T_1$



Initial condition: $v_x = V_{TL}$ and $v_o = V_H$

The output v_o will remain at same value until $v_o = v_x$ crosses the voltage V_{TH} . $[v_+ > v_-, v_o = V_H]$

$$v_o = v_x(t) = V_H + (V_{TL} - V_H) \exp\left(-\frac{t-0}{R_x C_x}\right)$$

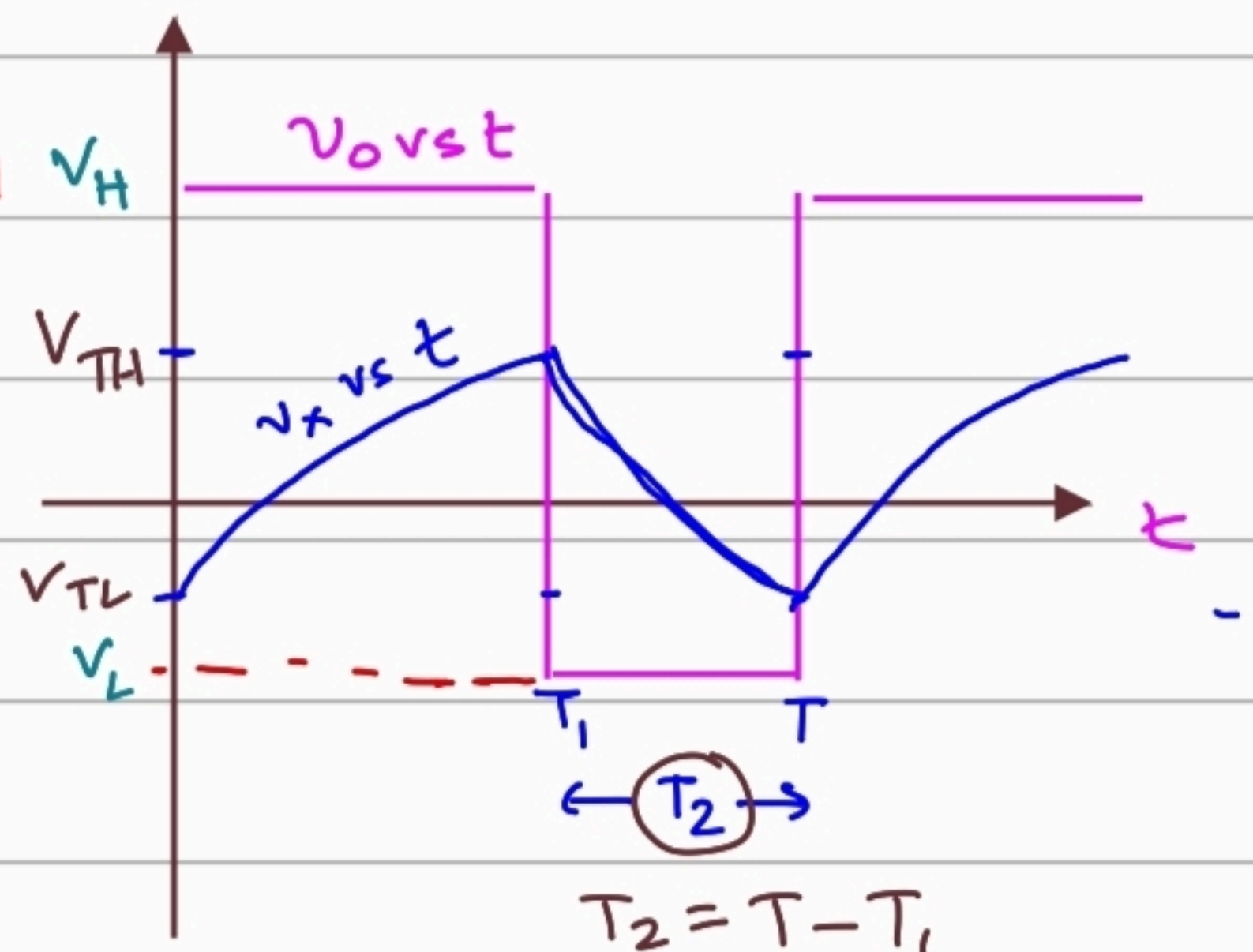
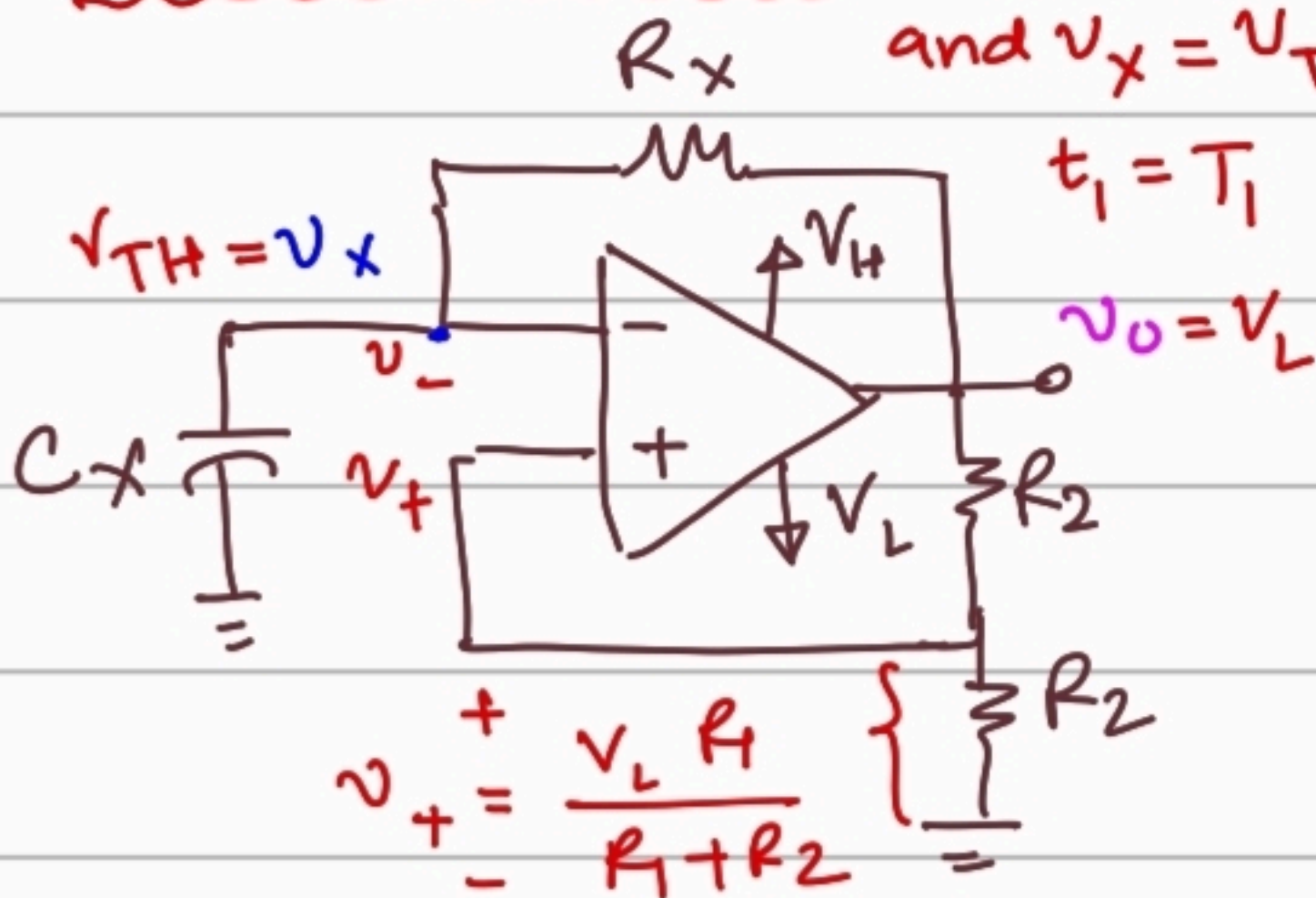
At $t = T_1$, $v_x(T_1) = V_{TH}$. Using this condition we can determine the value of T_1

$$v_x(T_1) = V_{TH} = V_H + (V_{TL} - V_H) \exp\left(-\frac{T_1}{R_x C_x}\right)$$

$$\Rightarrow T_1 = R_x C_x \ln \left| \frac{V_H - V_{TL}}{V_H - V_{TH}} \right|$$

② Case 2: $T_1 < t < T$

Initial condition: $v_o = V_L$ and $v_x = V_{TH}$



$(v_+ < v_-)$ Now output will remain at $v_o = V_L$ until v_x crosses V_{TL} .

$$v_c = v_x(t) = V_L + (V_{TH} - V_L) \exp\left(-\frac{t - T_1}{R_x C_x}\right)$$

At $t = T$ time $v_x(T) = V_{TL}$. So using this we find.

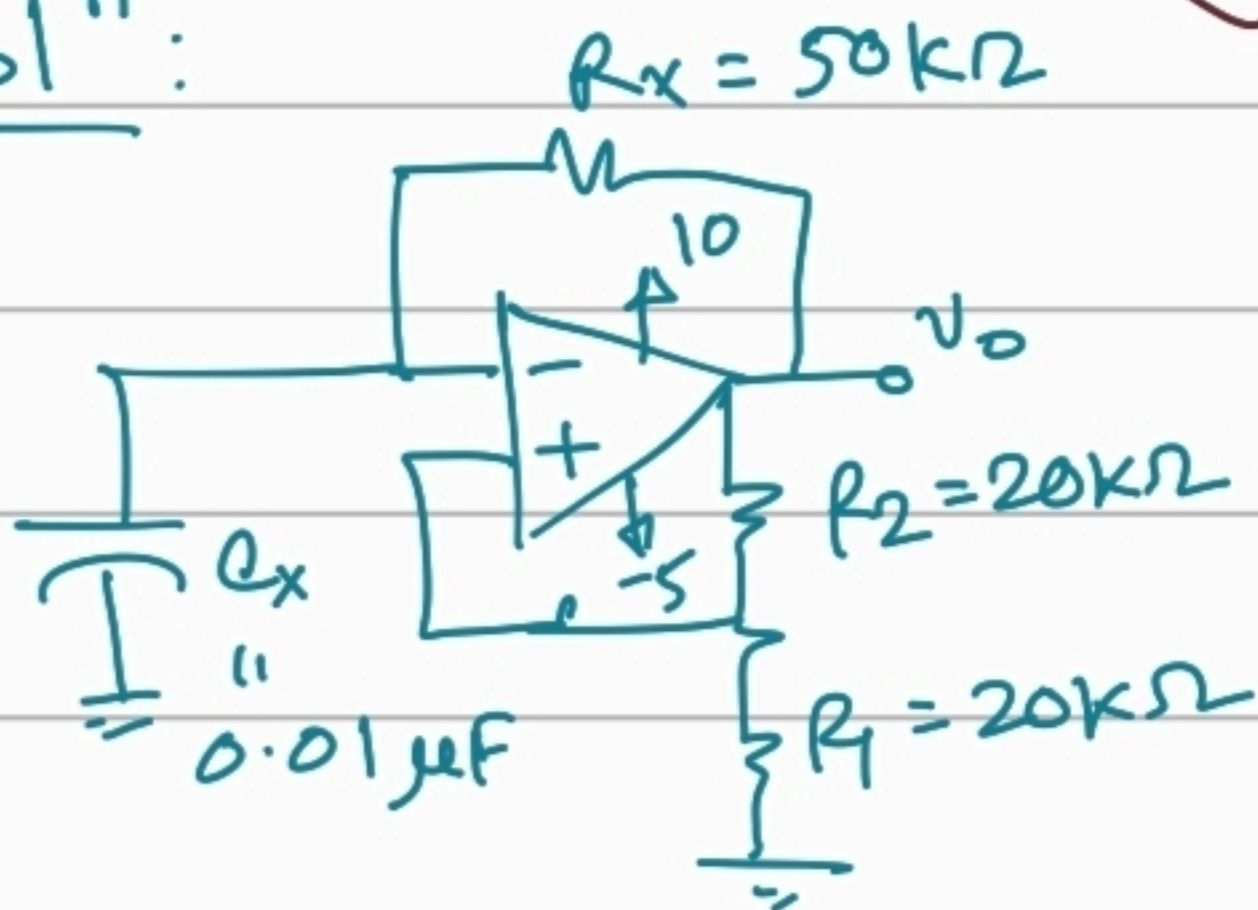
$$V_x(T) = V_{TL} = V_L + (V_{TH} - V_L) \exp\left(-\frac{(T - T_1)}{R_x C_x}\right)^{T_2}$$

$$\Rightarrow T_2 = R_x C_x \ln \left| \frac{V_L - V_{TH}}{V_L - V_{TL}} \right| \leftarrow \text{Just swap } H \leftrightarrow L \text{ to get } T_1.$$

Neamann's Exercise 15.8. For the Schmitt trigger

Oscillator, the saturation output voltages are $+10V$ and $-5V$. $R_1 = R_2 = 20k\Omega$, $R_x = 50k\Omega$ and $C_x = 0.01\mu F$. Determine the frequency and duty cycle.

Solⁿ:



$$V_{TH} = \frac{V_H R_1}{R_1 + R_2} = \frac{10 \times 20}{20 + 20} = 5V$$

$$V_{TL} = \frac{V_L R_1}{R_1 + R_2} = \frac{-5 \times 20}{20 + 20} = -2.5V$$

$$\begin{aligned} \tau = R_x C_x &= 50 \times 10^3 \times 0.01 \times 10^{-6} \Omega F \\ &= 0.5 \times 10^{-3} s \\ &= 0.5 ms \end{aligned}$$

Time duration of high output, $T_1 = \tau \ln \left| \frac{V_H - V_{TL}}{V_H - V_{TH}} \right|$

$$T_1 = 0.5 ms \times \ln \left| \frac{10 - (-2.5)}{10 - 5} \right| = 0.5 \ln(2.5) ms$$

Time duration of low output, $T_2 = \tau \ln \left| \frac{V_L - V_{TH}}{V_L - V_{TL}} \right|$

$$T_2 = 0.5 ms \times \ln \left| \frac{-5 - 5}{-5 - (-2.5)} \right| = 0.5 \ln(4) ms$$

$$\begin{aligned} \text{Total time period, } T &= T_1 + T_2 = 0.5 \ln(2.5 \times 4) \\ &= 0.5 \ln(10) \end{aligned}$$

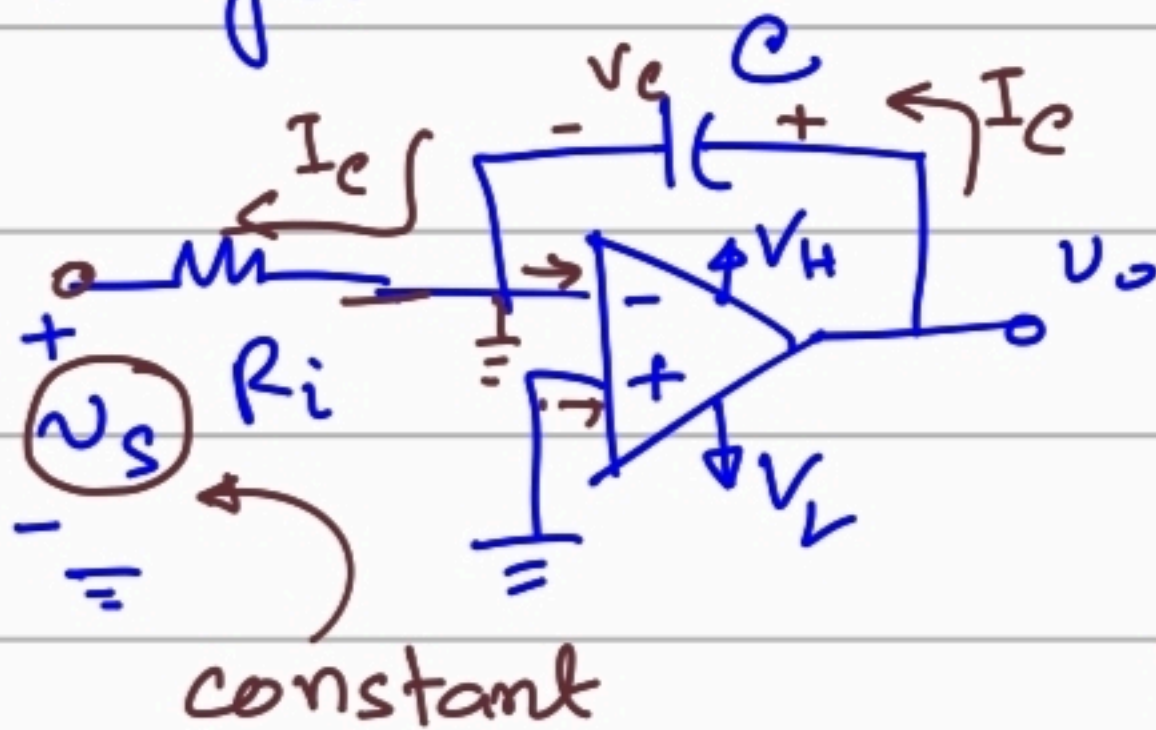
frequency. $f = \frac{1}{T} = \frac{1}{0.5 \ln(10) \text{ ms}} = 0.868 \text{ kHz}$
 $= \boxed{868 \text{ Hz}}$

Duty cycle = % of time output voltage is high with a time period

$$= \frac{T_1}{T} \times 100\% = \frac{\ln(2.5)}{\ln(10)} \times 100\% = \boxed{39.1\%}$$

Triangular Wave Generator:

Integrator:



We are assuming virtual ground is present here.

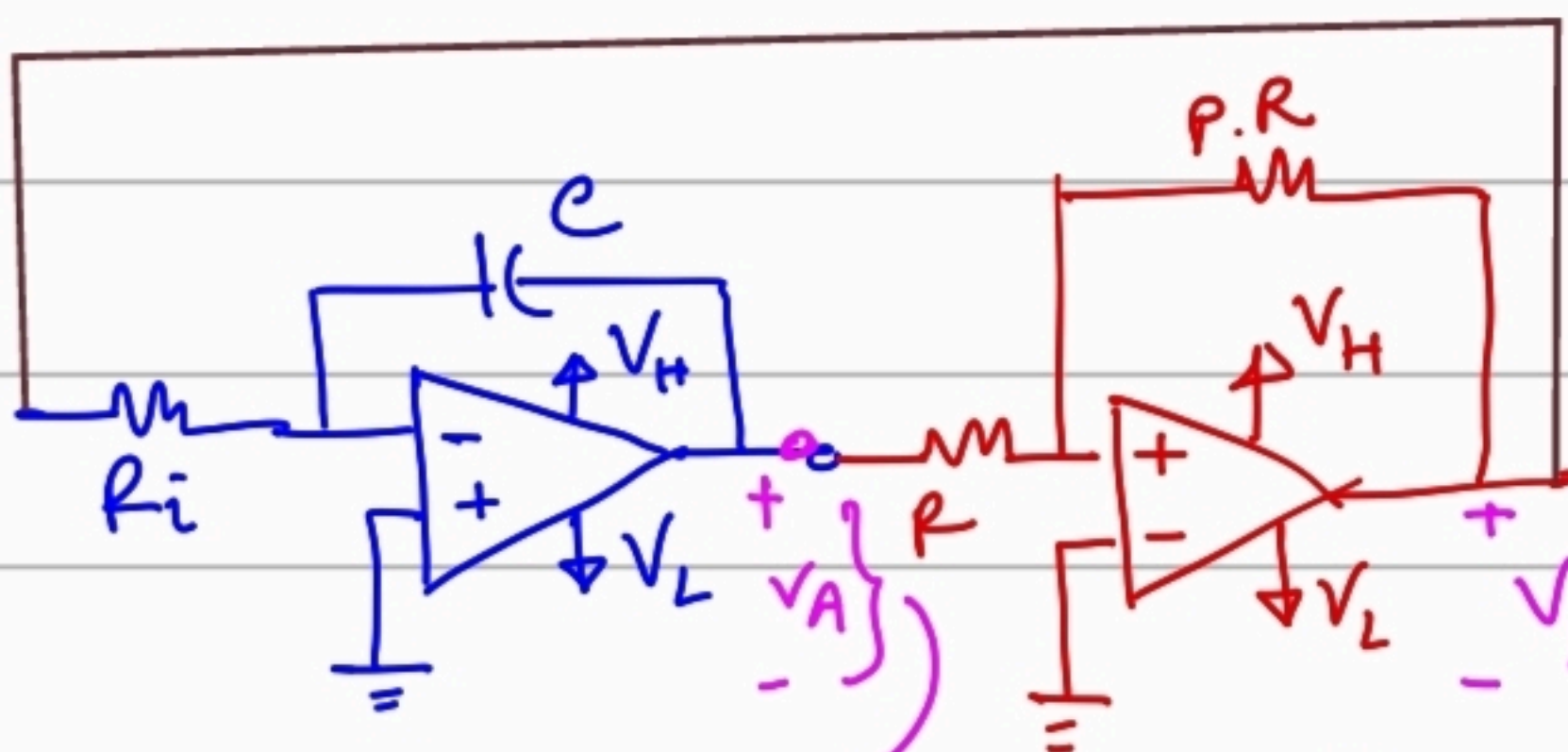
$$I_c = C \frac{dv_c}{dt} = \frac{0 - v_s}{R_i}$$

$$\Rightarrow \int_{v_c(t_1)}^{v_c(t)} dv_c = - \frac{v_s}{R_i C} \int_{t_1}^t dt$$

$$\Rightarrow v_c(t) - v_c(t_1) = - \frac{v_s}{R_i C} (t - t_1)$$

$$\Rightarrow \boxed{v_c(t) = v_c(t_1) - \frac{v_s}{R_i C} (t - t_1)}$$

$$v_c(t) = V_{\text{initial}}$$



Integrator

Non-inverting Schmitt trigger

Triangular wave

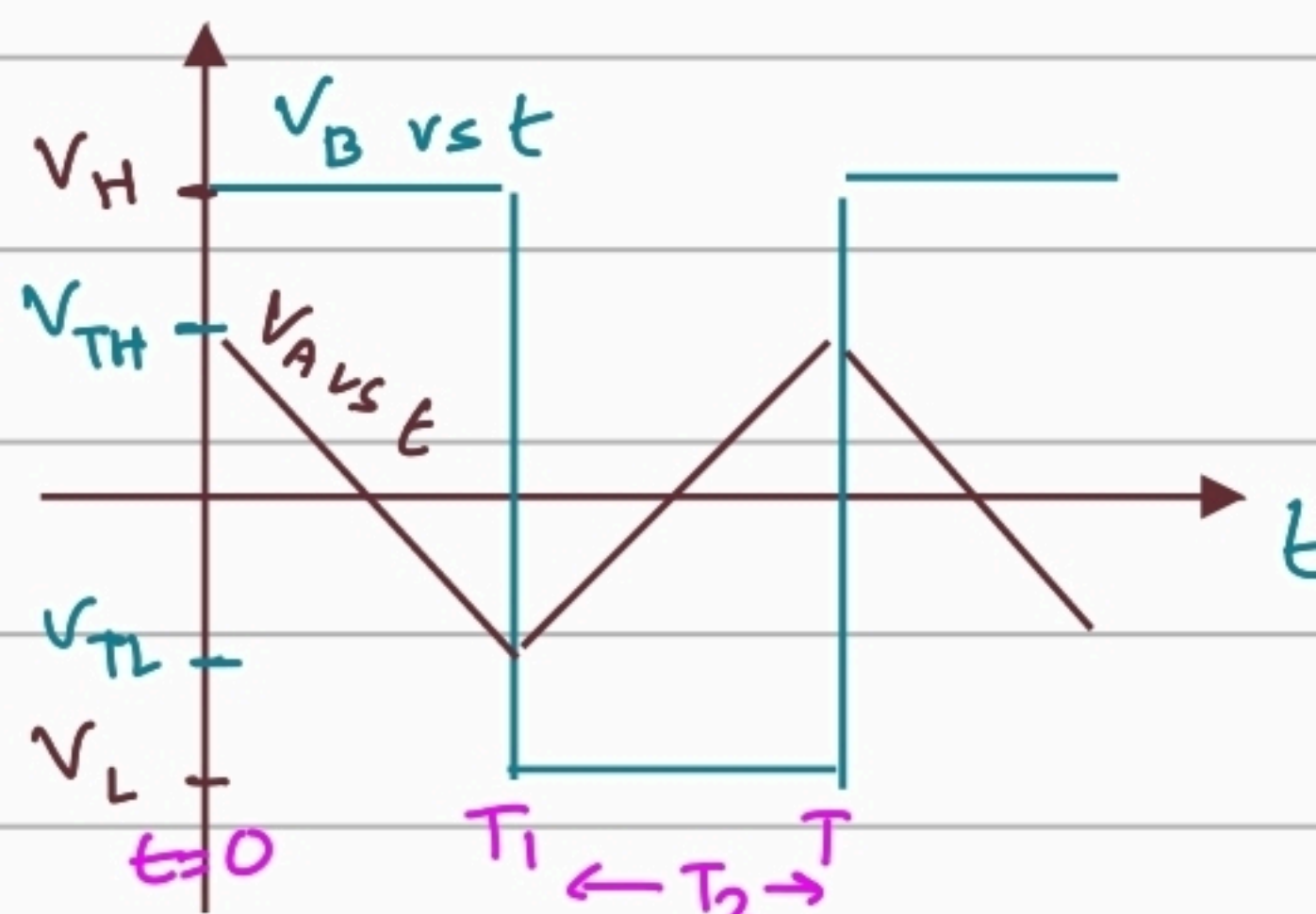
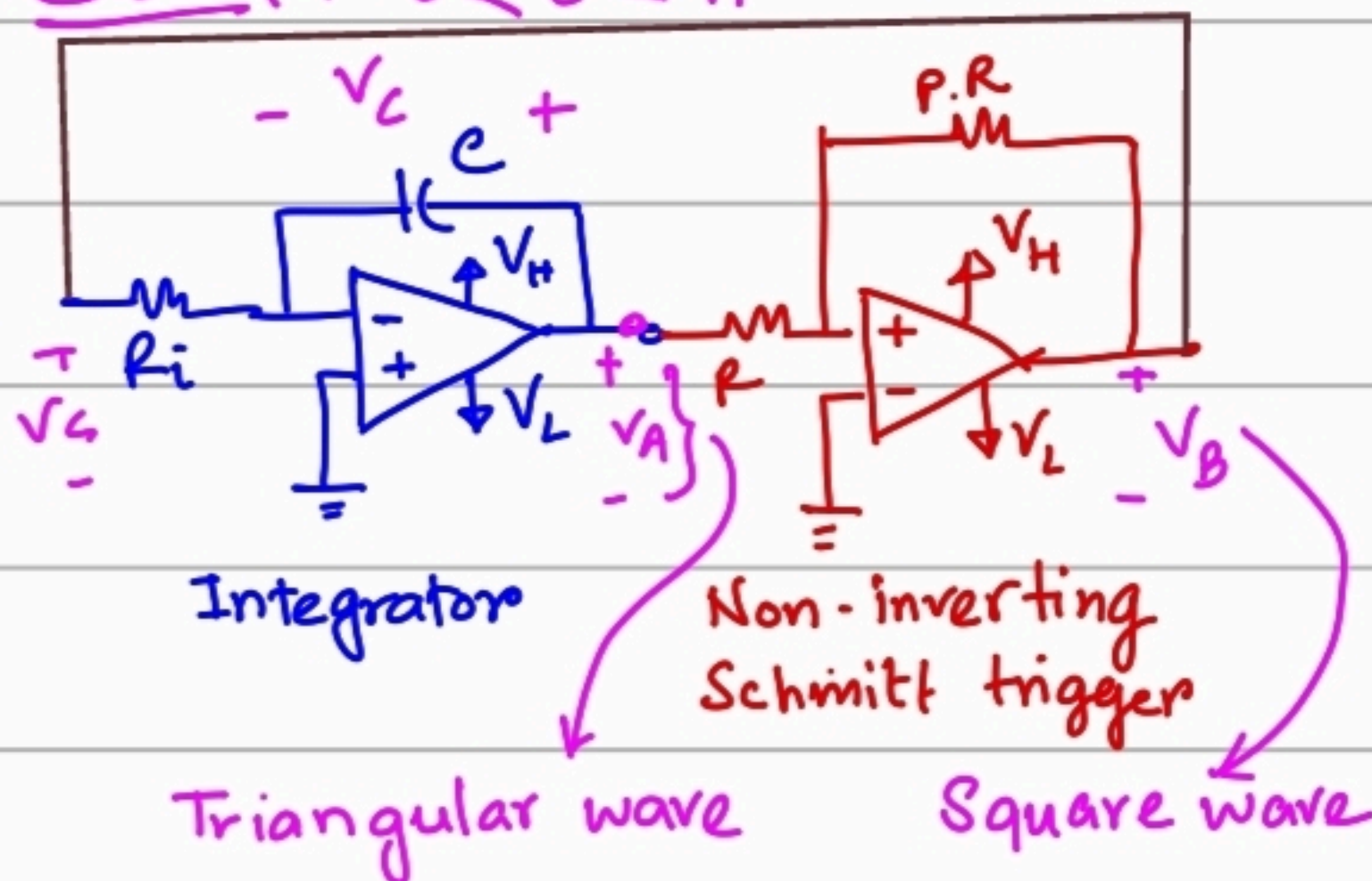
Square wave

$$V_{TL} = V_H \left(-\frac{R}{PR} \right)$$

$$V_{TL} = - \frac{V_H}{P}$$

$$V_{TH} = - \frac{V_L}{P}$$

Case 1: $0 < t < T_1$



Initial condition: $V_B = V_H$ and $V_A = V_{TH}$, $t_{initial} = 0$

$$V_C(t) = V_A(t) = V_{initial} - \frac{V_S}{R_i C} (t - t_{initial}) \quad [V_S = V_B = V_H]$$

$$\Rightarrow V_A(t) = V_{TH} - \frac{V_H}{R_i C} (t - 0)$$

Because $V_A(T_1) = V_{TL}$, we can find.

$$T_1 = R_i C \left(\frac{V_{TH} - V_{TL}}{V_H} \right)$$

Case 2: $T_1 < t < T$. Using similar analysis like square wave we can derive,

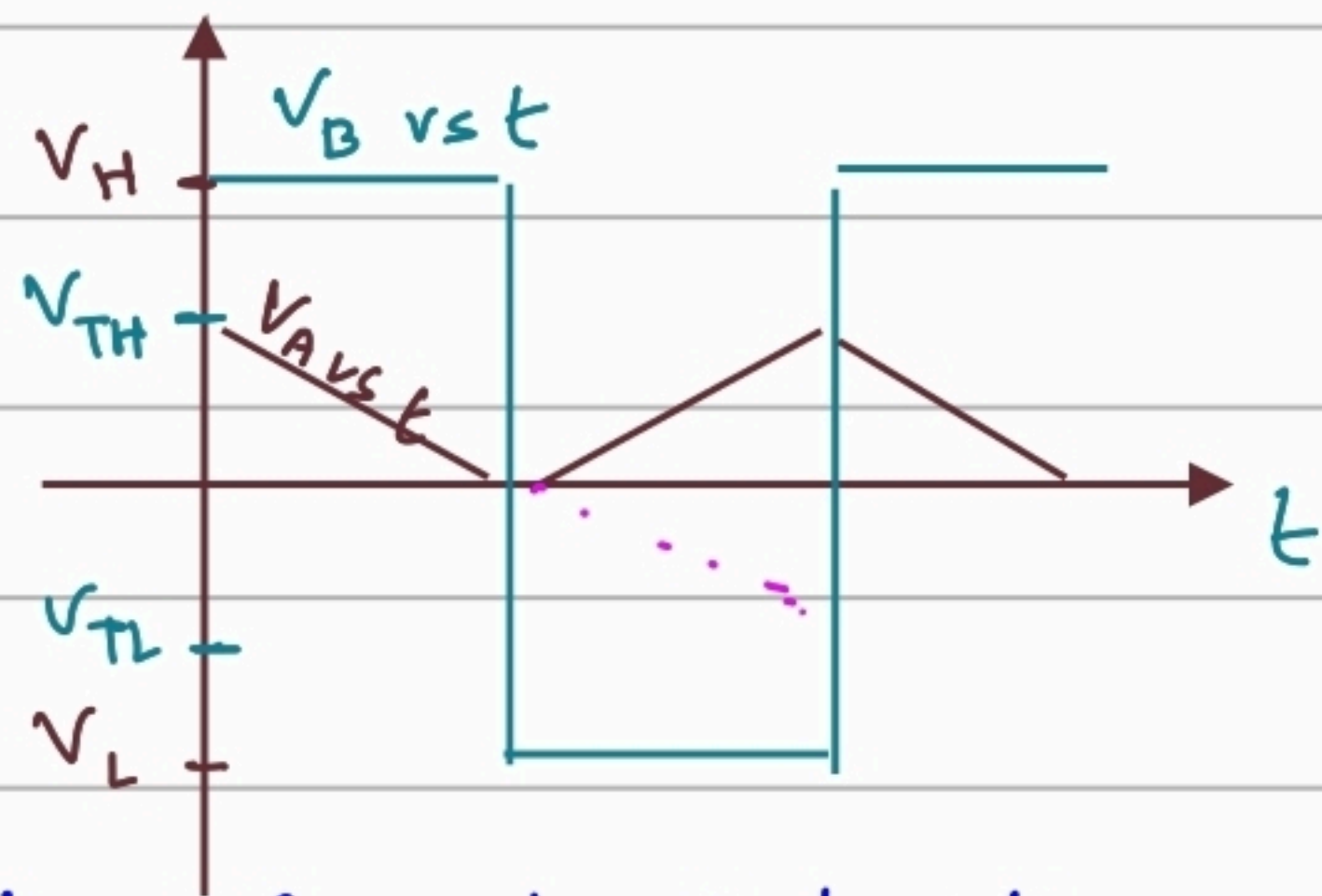
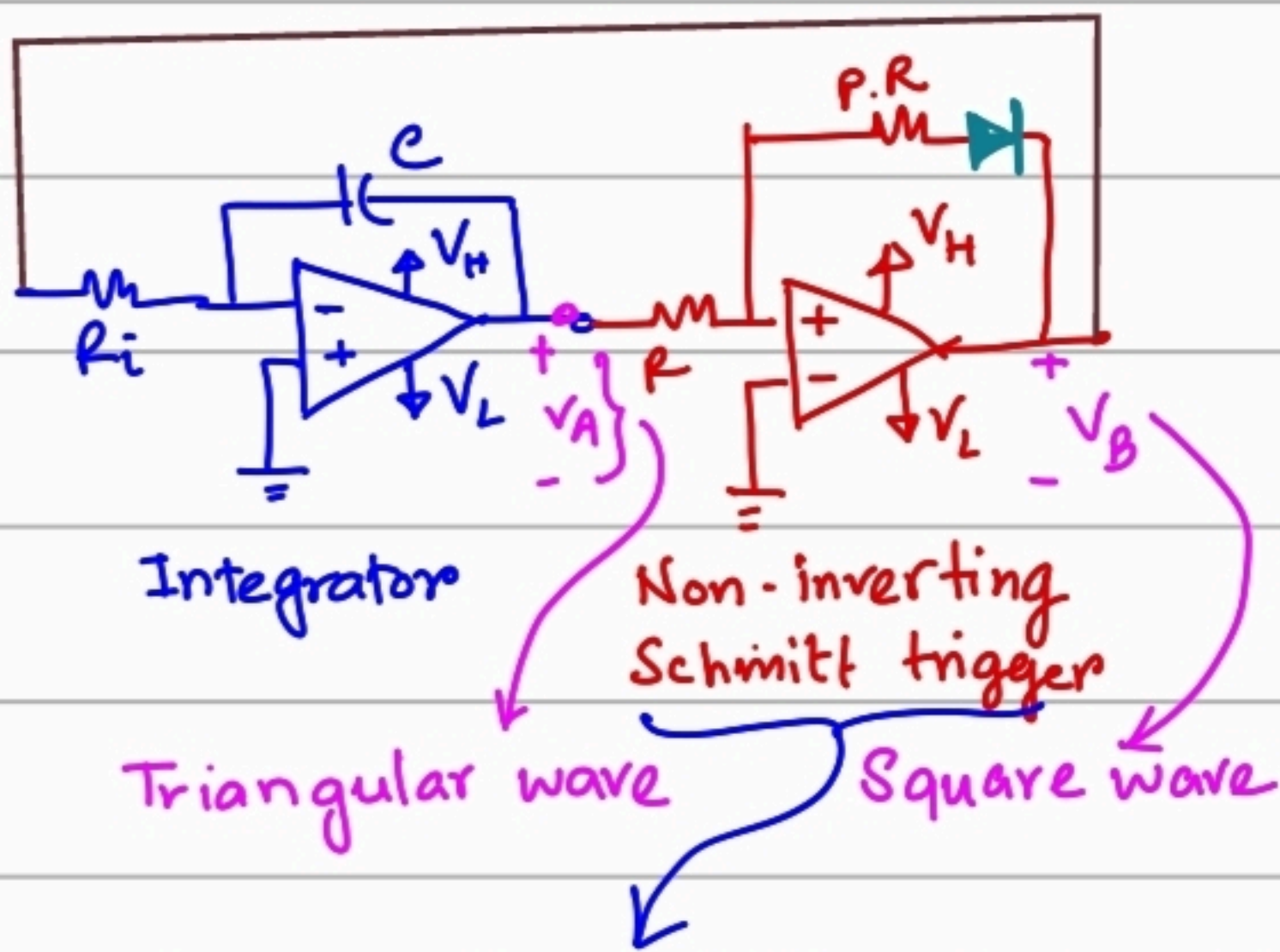
$$T_2 = R_i C \left(\frac{V_{TL} - V_{TH}}{V_L} \right) \quad H \leftrightarrow L$$

Total time period, $T = T_1 + T_2$. frequency, $f = \frac{1}{T}$

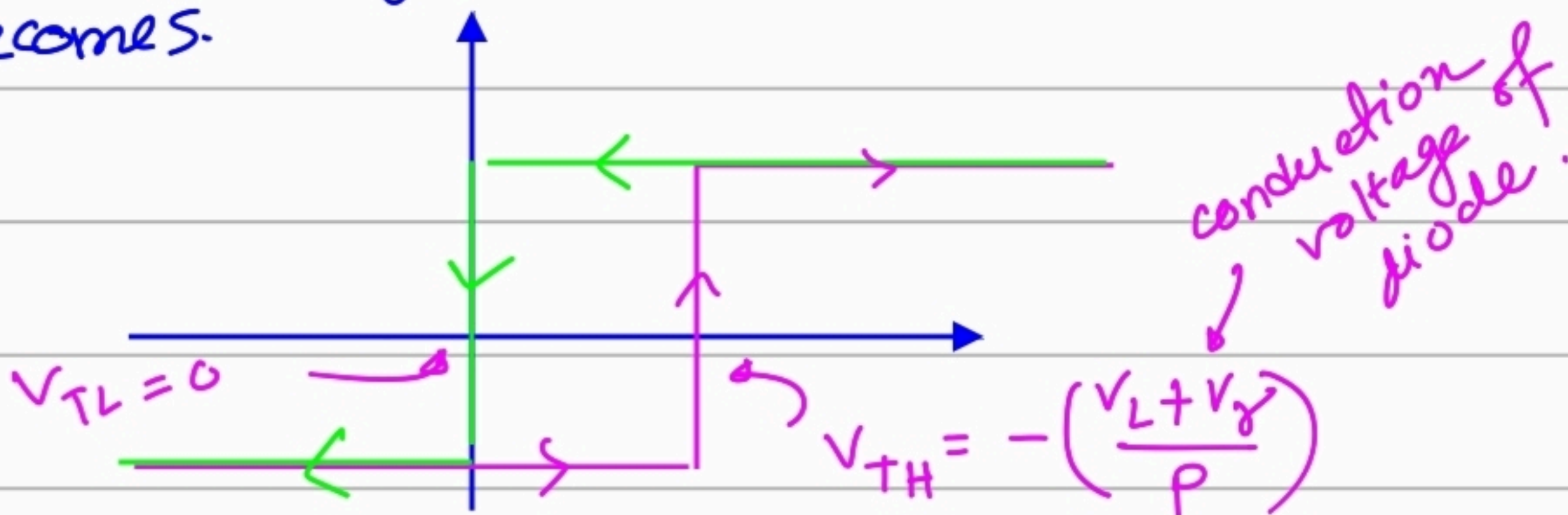
Special case: $V_L = -V_H$, $V_{TH} = -\frac{V_L}{P}$, $V_{TL} = -\frac{V_H}{P}$

$$f = \frac{P}{4R_i C}$$

Unipolar Triangular Wave Generator



After putting this diode the transfer characteristics becomes.



Assuming. $V_H = -V_L$, we will get

$$f = \frac{P}{2RC}$$